

TWO-STEP DYNAMICS AND FLAGGED FIBRES OF PREVIOUS-SMALLER DISTANCES

ANONYMOUS

ABSTRACT. Recomputing strict previous-smaller distances on an inversion sequence reaches an explicit recurrent involution after at most two updates. The nearest-smaller statistic is classical, but feedback compares the new distances themselves rather than following the original predecessor links. We prove that zero positions remain fixed and that the second iterate records precisely the original ascent set inside each positive block. At local block position j , the recurrent choices are 1 and j , which are exchanged by the next update. The maximum transient is two for every length at least four, with separate smaller-length boundaries. The recurrent and fixed populations are F_{2n-1} and F_{n+1} . Independently, we evaluate every target fibre at every time $t \geq 2$ by an alternating product of binomial coefficients. The formula retains the global position bounds of each block and distinguishes the two eventual phases. Complete proofs, together with bounded exact verification, separate this endpoint-to-fibre description from its classical static nearest-smaller and inversion-statistic ingredients.

1. THE MAP AND ITS SCOPE

For $n \geq 1$ use indices $0, \dots, n-1$ and the inversion-sequence box

$$(1) \quad E_n = \{x = (x_0, \dots, x_{n-1}) : 0 \leq x_i \leq i\}, \quad |E_n| = n!.$$

The strict previous-smaller distance map is

$$(2) \quad P(x)_i = \begin{cases} i - \max\{h < i : x_h < x_i\}, & \text{if this set is nonempty,} \\ 0, & \text{otherwise.} \end{cases}$$

It is a self-map of E_n : a positive distance lies between 1 and i . Every output is computed from the same old word. In particular, the update recomputes comparisons; it does not follow a pointer stored in x_i . The tail $\tau(x)$ is the least $t \geq 0$ at which $P^t(x)$ is periodic.

Static nearest-smaller values are classical algorithmic objects [1]. Parent-distance representations also encode Cartesian trees [3], although that paper's convention uses an earlier value at most the current value. Ties matter here: (2) sends 011 to 012, whereas the weak convention sends it to 011. Neither the static search nor its data structures are contributions of this note.

The two results below concern the feedback itself. First, its second iterate decodes the original zero positions and internal ascent sets, and the resulting core has an explicit involution. Second, each target in either eventual phase has an evaluated source count with its original position-dependent bounds. Ascent sets and zero positions are already classical joint statistics: under the usual inversion code they correspond to permutation descents and left-to-right maxima [2]. The contribution asserted here is the concrete iterate and its fibre evaluation, not the discovery of those statistics or of the counting tools.

2. TWO-STEP DECODING AND THE RECURRENT INVOLUTION

Zeros are invariant under P . If $x_i = 0$, no nonnegative entry is smaller; if $x_i > 0$, the earlier entry $x_0 = 0$ is smaller. Let $\mathcal{B}(x)$ be the set of pairs (r, m) such that $r+1, \dots, r+m$ is a maximal positive block of x , immediately following the zero at position r . Its output distances lie in $\{1, \dots, j\}$ at position $r+j$, because the nearest smaller predecessor cannot be to the left of r .

Lemma 2.1 (First-image inequality). *Let $(r, m) \in \mathcal{B}(x)$ and $b_j = P(x)_{r+j}$. Then $b_1 = 1$, and for $2 \leq j \leq m$ either $b_j = 1$ or $b_j > b_{j-1}$.*

Proof. Write p_j for the nearest smaller predecessor of x_{r+j} . If $b_j > 1$, then $x_{r+j} \leq x_{r+j-1}$. Every entry strictly between p_{j-1} and $r+j-1$ is at least x_{r+j-1} , hence at least x_{r+j} ; the entry at $r+j-1$ is not smaller either. Thus $p_j \leq p_{j-1}$ and $b_j = r+j-p_j \geq r+j-p_{j-1} = b_{j-1}+1$. The first entry of the block has its zero barrier as nearest smaller predecessor, so $b_1 = 1$. $\square$

Define the block core

$$(3) \quad \mathcal{C}_n = \{y \in E_n : y_{r+j} \in \{1, j\} \text{ for all } (r, m) \in \mathcal{B}(y), 1 \leq j \leq m\}.$$

On this set let J keep zeros and block starts fixed and exchange $1 \leftrightarrow j$ at each local position $j \geq 2$.

Theorem 2.2 (Exact endpoint and sharp height). *The zero set of $P^2(x)$ is that of x . In each positive block,*

$$(4) \quad P^2(x)_{r+1} = 1, \quad P^2(x)_{r+j} = \begin{cases} j, & x_{r+j-1} < x_{r+j}, \\ 1, & x_{r+j-1} \geq x_{r+j}, \end{cases} \quad 2 \leq j \leq m.$$

Moreover $P^2(E_n) = \mathcal{C}_n$, $P|_{\mathcal{C}_n} = J$, and $P^4 = P^2$. The recurrent set is exactly $\mathcal{C}_n$, all periods divide two, and

$$(5) \quad \max_{x \in E_n} \tau(x) = \begin{cases} 0, & n \leq 2, \\ 1, & n = 3, \\ 2, & n \geq 4. \end{cases}$$

Proof. For $j \geq 2$, $b_j = 1$ is equivalent to the original adjacent ascent in (4). When $b_j = 1$, its only smaller value in the block or left barrier is zero, so the next distance is j . When $b_j > 1$, Lemma 2.1 makes the immediately preceding distance smaller, so the next distance is one. This proves (4) and $P^2(E_n) \subseteq \mathcal{C}_n$.

At a core coordinate with value $j \geq 2$, the preceding core value is at most $j - 1$, and the next distance is one. At a core coordinate with value one, only the left zero is smaller, and the next distance is j . Thus P restricts to J , with J^2 the identity. Each core word is therefore its own two-step preimage. This proves the exact second image and $P^4 = P^2$. Every periodic state lies in the second image, so the recurrent set is precisely the core.

For $n = 1, 2$ the whole carrier is fixed. At $n = 3$ the six states are 000, 001, 002, 010, 011, 012; only 002 is outside the core, and it maps to 001. For $n \geq 4$, prepend $n - 4$ zeros to 0122. Its first-image positive block is 112, whose last value is neither one nor its local index three. That image is not recurrent, proving the sharp lower bound two; the second-image theorem supplies the upper bound. $\square$

Corollary 2.3 (Core census). *Let $F_0 = 0, F_1 = 1$ and $F_{k+2} = F_{k+1} + F_k$. There are F_{2n-1} recurrent states, F_{n+1} fixed states, and $(F_{2n-1} - F_{n+1})/2$ strict two-cycles.*

Proof. For the $N = n - 1$ positions after the initial zero, let a_N, b_N count core words ending respectively in zero and in a positive entry, with $(a_0, b_0) = (1, 0)$. A new zero has one choice, a positive block starts with one, and its continuation has the two distinct choices 1, j . Hence

$$(6) \quad (a_{N+1}, b_{N+1}) = (a_N + b_N, a_N + 2b_N).$$

For $N \geq 1$, induction gives $(a_N, b_N) = (F_{2N-1}, F_{2N})$, so the total is F_{2N+1} , also valid at $N = 0$. A core word is fixed exactly when every positive block is a singleton. Binary words of length N with no consecutive ones are counted by F_{N+2} : conditioning on an initial zero or an initial 10 gives the Fibonacci recurrence, with initial counts 1, 2. The remaining core states are paired by J . $\square$

3. EVERY TARGET FIBRE AFTER TWO STEPS

Fix (r, m) with $r \geq 0, m \geq 1$ and a set $A \subseteq \{2, \dots, m\}$. For $B \subseteq A$, let $\text{Seg}(B)$ partition $\{1, \dots, m\}$ into consecutive segments by cutting immediately before each $j \in B$. Define the globally flagged count

$$(7) \quad D_{r,m}(A) = \sum_{B \subseteq A} (-1)^{|A|-|B|} \prod_{[a,b] \in \text{Seg}(B)} \binom{r+b}{b-a+1}.$$

The word “flagged” refers to the position-dependent bounds $r + j$; the offset r is part of the count, not an ignorable block label. For $y \in \mathcal{C}_n$ and $t \geq 2$, set

$$(8) \quad A_t(y; r, m) = \begin{cases} \{j \in \{2, \dots, m\} : y_{r+j} = j\}, & t \text{ even}, \\ \{j \in \{2, \dots, m\} : y_{r+j} = 1\}, & t \text{ odd}. \end{cases}$$

Theorem 3.1 (Target-resolved post-collapse fibres). *For every $n \geq 1$, every $t \geq 2$ and every $y \in E_n$,*

$$(9) \quad |(P^t)^{-1}(y)| = \begin{cases} \prod_{(r,m) \in \mathcal{B}(y)} D_{r,m}(A_t(y; r, m)), & y \in \mathcal{C}_n, \\ 0, & y \notin \mathcal{C}_n. \end{cases}$$

The empty product is one. Thus the all-zero target has exactly one source.

Proof. Theorem 2.2 gives $P^t(x) = J^{t-2}P^2(x)$. A source of y must have exactly its zero set. For a core target, (4) says that the source's internal ascent set in block (r, m) must be $A_t(y; r, m)$, and this condition is sufficient. Different positive blocks are independent after the zero positions are fixed. It remains to count one block with entries $1 \leq x_{r+j} \leq r+j$ and exact internal ascent set A .

Impose nonascent on each edge outside A . Inclusion–exclusion on failures of the required ascents gives the sum over $B \subseteq A$ with sign $(-1)^{|A|-|B|}$, where ascents are allowed only before positions in B . The resulting word is weakly decreasing on every segment $[a, b]$ of $\text{Seg}(B)$. All entries of that segment are at most its first entry, which is at most $r+a$. Conversely, every weakly decreasing sequence of length $b-a+1$ in $\{1, \dots, r+a\}$ satisfies the later, increasing position bounds. Counting its value multiplicities gives

$$\binom{(r+a) + (b-a+1) - 1}{b-a+1} = \binom{r+b}{b-a+1}.$$

Segments are independent, giving (7); blocks are independent, giving (9). Noncore targets have no source because every iterate at time at least two lies in $\mathcal{C}_n$. $\square$

Example 3.2 (The global offset cannot be dropped). For a two-letter block, the number of strictly rising positive sources is

$$D_{r,2}(\{2\}) = (r+1)(r+2) - \binom{r+2}{2} = \binom{r+2}{2}.$$

It equals one at offset zero and three at offset one. Thus an unflagged ascent count cannot substitute for the block factor in (9). At even times this ascent factor belongs to the local target 12; at odd times at least three it belongs to the exchanged target 11.

4. VERIFICATION AND LIMITATIONS

The paper-local standard-library verifier exhausts E_n for $1 \leq n \leq 8$ (46,233 input states). It tests zero barriers, the first-image inequality, the endpoint, core action, sharp heights and both Fibonacci counts. Every target fibre is compared with literal iteration at $t = 2, 3, 4, 5$. A separate bounded source-word enumeration checks 315 flagged ascent patterns for $0 \leq r \leq 4$, $1 \leq m \leq 6$. The resulting 485,578 assertions are finite pressure on the proofs, not an argument for larger parameters.

The results concern the full box (1) and the strict rule (2). They do not characterize the first image or one-step fibres, classify largest fibres, or extend the carrier to arbitrary words. Under the classical inversion code, (9) also counts an intersection of prescribed permutation descent and record-position sets; that interpretation is not a new statistic [2]. Fibonacci recurrence, inclusion–exclusion and multiset counting likewise remain standard. The narrow conclusion is the exact two-step decoder and the explicit source count for each target and eventual phase. No claim of global priority or external publication clearance is made.

REFERENCES

- [1] Omer Berkman, Baruch Schieber, and Uzi Vishkin. Optimal doubly logarithmic parallel algorithms based on finding all nearest smaller values. *Journal of Algorithms*, 14(3):344–370, 1993. doi: 10.1006/jagm.1993.1018.
- [2] Zhicong Lin and Dongsu Kim. A sextuple equidistribution arising in pattern avoidance. *Journal of Combinatorial Theory, Series A*, 155:267–286, 2018. doi: 10.1016/j.jcta.2017.11.009.
- [3] Sung Gwan Park, Amihoud Amir, Gad M. Landau, and Kunsu Park. Cartesian tree matching and indexing. In *30th Annual Symposium on Combinatorial Pattern Matching (CPM 2019)*, volume 128 of *Leibniz International Proceedings in Informatics*, pages 16:1–16:14. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2019. doi: 10.4230/LIPIcs.CPM.2019.16.